



OPERATING SYSTEMS

UE24CS242B

IPC - Shared Memory & Message Passing

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Slides Credits for all the PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

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Interprocess Communication

- Processes within a system may be *independent* or *cooperating*
- Cooperating process can affect or be affected by other processes, including sharing data
- Reasons for cooperating processes:
 - Information sharing
 - Computation speedup
 - Modularity
 - Convenience
- Cooperating processes need **interprocess communication (IPC)**
- Two models of IPC
 - **Shared memory**
 - **Message passing**



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Cooperating Processes

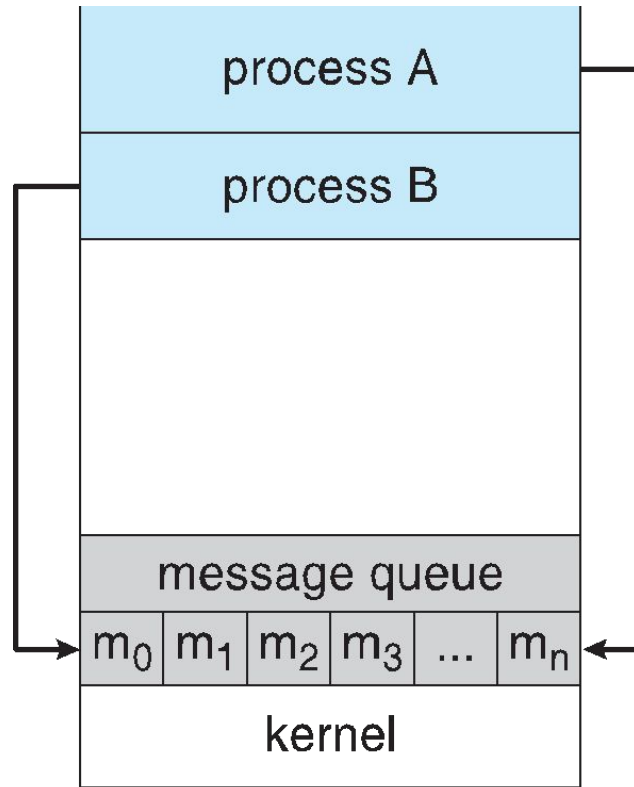
- ***Independent*** process cannot affect or be affected by the execution of another process
- ***Cooperating*** process can affect or be affected by the execution of another process
- Advantages of process cooperation
 - Information sharing
 - Computation speed-up
 - Modularity
 - Convenience



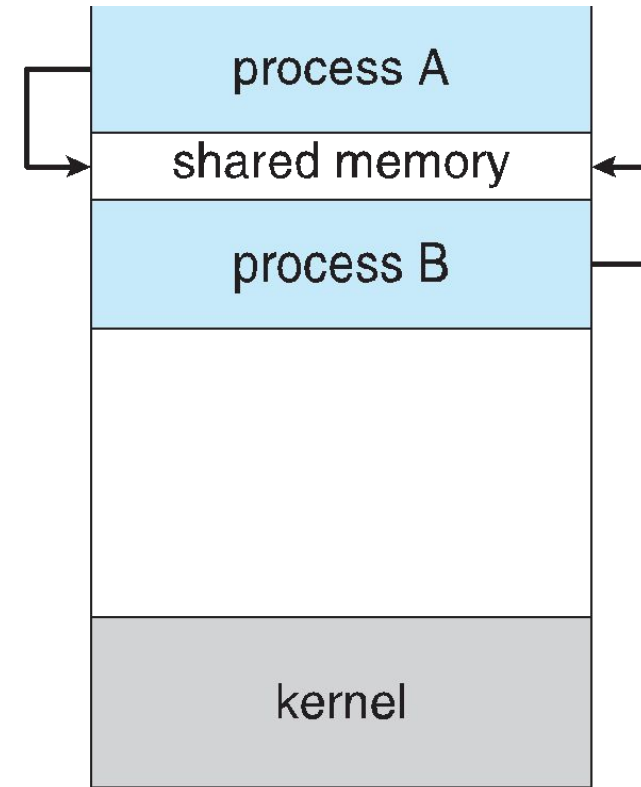
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Communication Models

- Two models of IPC
 - a) **Message passing**
 - b) **Shared memory**



(a)



(b)

- Paradigm for cooperating processes, *producer* process produces information that is consumed by a *consumer* process
 - **unbounded-buffer** places no practical limit on the size of the buffer
 - Consumer may have to wait for new items, but the producer can always produce new items
 - **bounded-buffer** assumes that there is a fixed buffer size
 - Consumer must wait if the buffer is empty; producer must wait if the buffer is full

- Shared data

```
#define BUFFER_SIZE 10
typedef struct {
    ...
} item;

item buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
```
- Shared buffer is implemented as a circular array with 2 logical pointers: **in** and **out**
- Buffer is empty when **in == out**; buffer is full when $((\text{in} + 1) \% \text{BUFFER_SIZE}) == \text{out}$
- Variable **in** points to the next free position in the buffer; **out** points to the first full position in the buffer
- Solution is correct, but can only use $\text{BUFFER_SIZE}-1$ elements


```
item next_produced;
while (true) {
    /* produce an item in next_produced */
    while (((in + 1) % BUFFER_SIZE) == out)
        ; /* do nothing */
    buffer[in] = next_produced;
    in = (in + 1) % BUFFER_SIZE;
}
```

```
item next_consumed;
while (true) {
    while (in == out)
        ; /* do nothing */
    next_consumed = buffer[out];
    out = (out + 1) % BUFFER_SIZE;

    /* consume the item in next_consumed
*/
}
```

```
item next_consumed;
while (true) {
    while (in == out)
        ; /* do nothing */
    next_consumed = buffer[out];
    out = (out + 1) % BUFFER_SIZE;

    /* consume the item in next_consumed */
}
```

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Interprocess Communication – Shared Memory

- An area of memory shared among the processes that wish to communicate
- The communication is under the control of the users processes not the operating system.
- Major issues is to provide mechanism that will allow the user processes to synchronize their actions when they access shared memory.

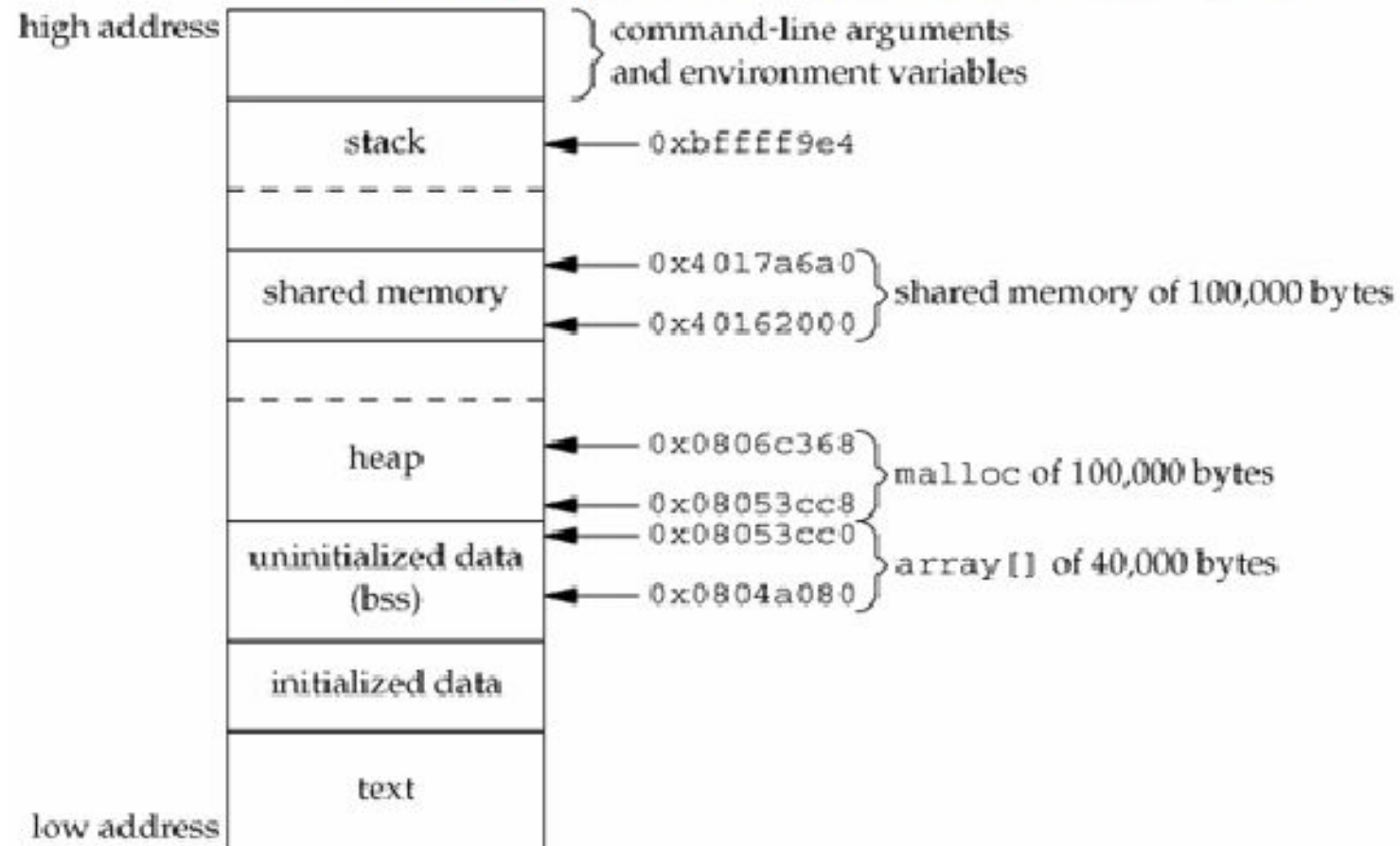


- Shared memory allows two or more processes to share a given region of memory.
- Shared memory is the fastest form of IPC, because the data does not need to be copied between the client and the server.
- The only trick in using shared memory is synchronizing access to a given region among multiple processes.
- If the server is placing data into a shared memory region, the client shouldn't try to access the data until the server is done.
- Often, semaphores are used to synchronize shared memory access. (record locking can also be used.)

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Shared Memory (Cont.)

Memory layout on an Intel-based Linux system



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Interprocess Communication – Message Passing

- Mechanism for processes to communicate and to synchronize their actions
- Message system – processes communicate with each other without resorting to shared variables
- IPC facility provides two operations:
 - **send**(*message*)
 - **receive**(*message*)
- The *message* size is either fixed or variable



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Message Passing (Cont.)

- If processes P and Q wish to communicate, they need to:
 - Establish a ***communication link*** between them
 - Exchange messages via send/receive
- Implementation issues:
 - How are links established?
 - Can a link be associated with more than two processes?
 - How many links can there be between every pair of communicating processes?
 - What is the capacity of a link?
 - Is the size of a message that the link can accommodate fixed or variable?
 - Is a link unidirectional or bi-directional?



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Message Passing (Cont.)

- Implementation of communication link
 - Physical:
 - Shared memory
 - Hardware bus
 - Network
 - Logical:
 - Direct or indirect
 - Synchronous or asynchronous
 - Automatic or explicit buffering



- Processes must name each other explicitly:
 - **send** ($P, message$) – send a message to process P
 - **receive**($Q, message$) – receive a message from process Q
- Properties of communication link
 - Links are established automatically
 - A link is associated with exactly one pair of communicating processes
 - Between each pair there exists exactly one link
 - The link may be unidirectional, but is usually bi-directional

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Indirect Communication

- Messages are directed and received from mailboxes (also referred to as ports)
 - Each mailbox has a unique id
 - Processes can communicate only if they share a mailbox
- Properties of communication link
 - Link established only if processes share a common mailbox
 - A link may be associated with many processes
 - Each pair of processes may share several communication links
 - Link may be unidirectional or bi-directional



- Operations
 - create a new mailbox (port)
 - send and receive messages through mailbox
 - destroy a mailbox
- Primitives are defined as:
 - **send**(*A, message*) – send a message to mailbox A
 - **receive**(*A, message*) – receive a message from mailbox A

- Mailbox sharing
 - P_1 , P_2 and P_3 share mailbox A
 - P_1 sends; P_2 and P_3 receive
 - Who gets the message?
- Solutions
 - Allow a link to be associated with at most two processes
 - Allow only one process at a time to execute a receive operation
 - Allow the system to select arbitrarily the receiver. Sender is notified who the receiver was.

- Message passing may be either blocking or non-blocking
- **Blocking** is considered **synchronous**
 - **Blocking send** -- the sender is blocked until the message is received
 - **Blocking receive** -- the receiver is blocked until a message is available
- **Non-blocking** is considered **asynchronous**
 - **Non-blocking send** -- the sender sends the message and continue
 - **Non-blocking receive** -- the receiver receives:
 - A valid message, or
 - Null message
- Different combinations possible
 - If both send and receive are blocking, we have a **rendezvous** between the sender and the receiver

- Producer-consumer becomes trivial

```
message next_produced;  
while (true) {  
    /* produce an item in next_produced */  
    send(next_produced);  
}
```

```
message next_consumed;  
while (true) {  
    receive(next_consumed);  
  
    /* consume the item in next_consumed */  
}
```

- Queue of messages attached to the link (direct or indirect); messages reside in a temporary queue.
- Queues can be implemented in one of three ways
 1. Zero capacity – no messages are queued on a link.
Sender must wait for receiver (rendezvous)
 2. Bounded capacity – finite length of n messages
Sender must wait if link full
 3. Unbounded capacity – infinite length
Sender never waits
- Zero-capacity case is sometimes referred to as a message system with no buffering; other cases are referred to as systems with automatic buffering



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System calls for Inter Process Communication

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System calls for Inter Process Communication

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- Pipes are the oldest form of UNIX System IPC and are provided by all UNIX systems.
- Pipes have two limitations.
 1. Historically, they have been half duplex (i.e., data flows in only one direction). Some systems now provide full-duplex pipes, but for maximum portability, we should never assume that this is the case.
 2. Pipes can be used only between processes that have a **common ancestor**.

Normally, a pipe is created by a process, that process calls fork, and the pipe is used between the parent and the child.

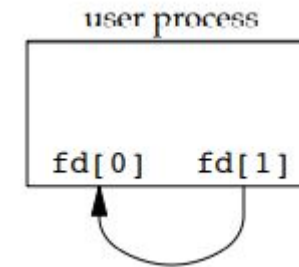
- A pipe is created by calling the **pipe()** function.

```
#include <unistd.h>
int pipe(int fd[2]);
```

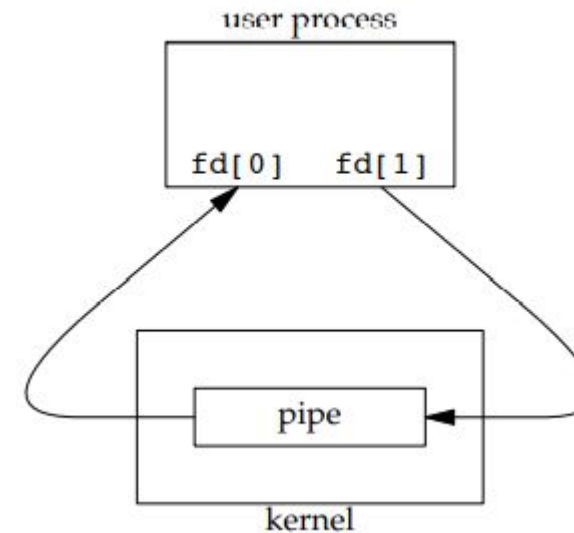
- Return value: 0 if OK, -1 on error
- Two file descriptors are returned through the fd argument:
 1. fd[0] is open for reading
 2. fd[1] is open for writing.
- The output of fd[1] is the input for fd[0]

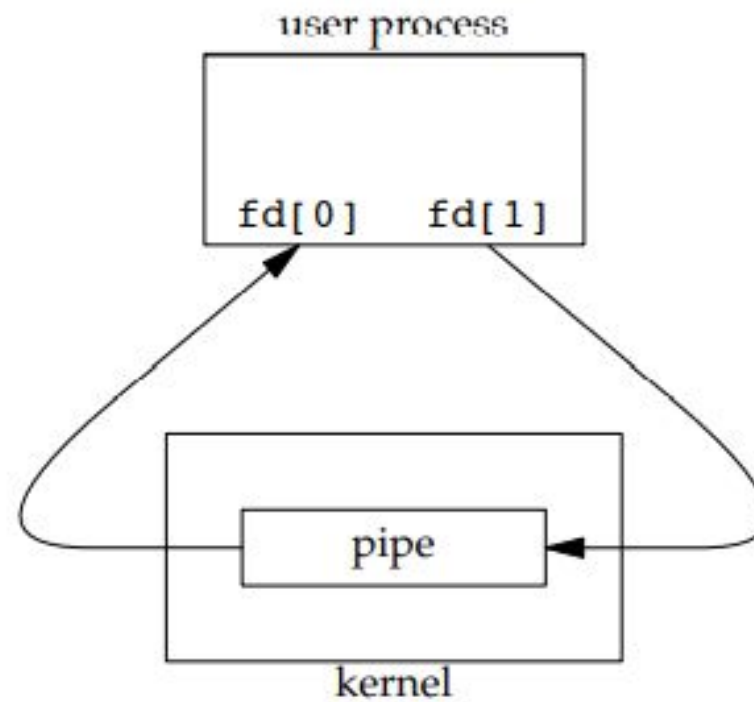
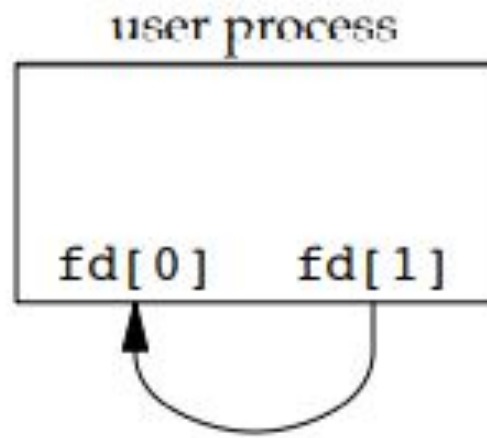
- Two ways to picture a half-duplex pipe

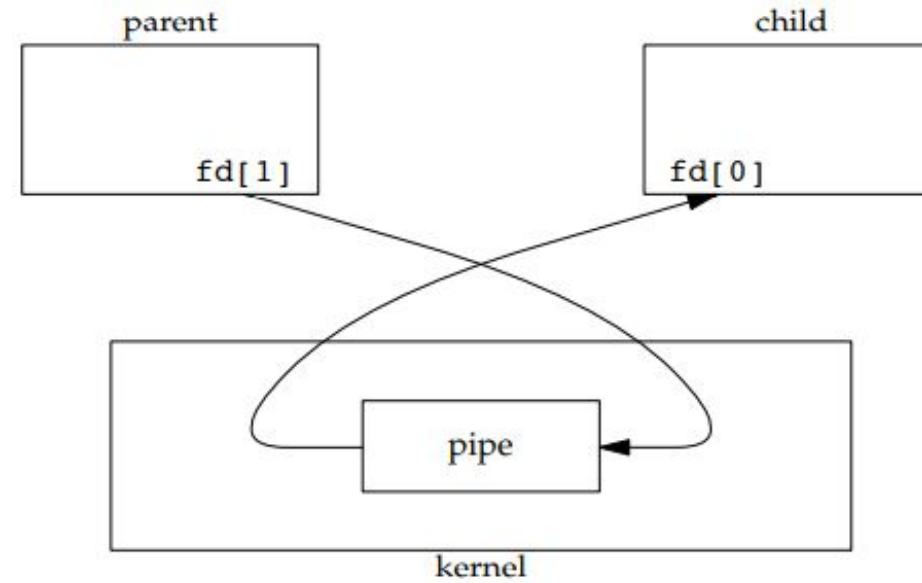
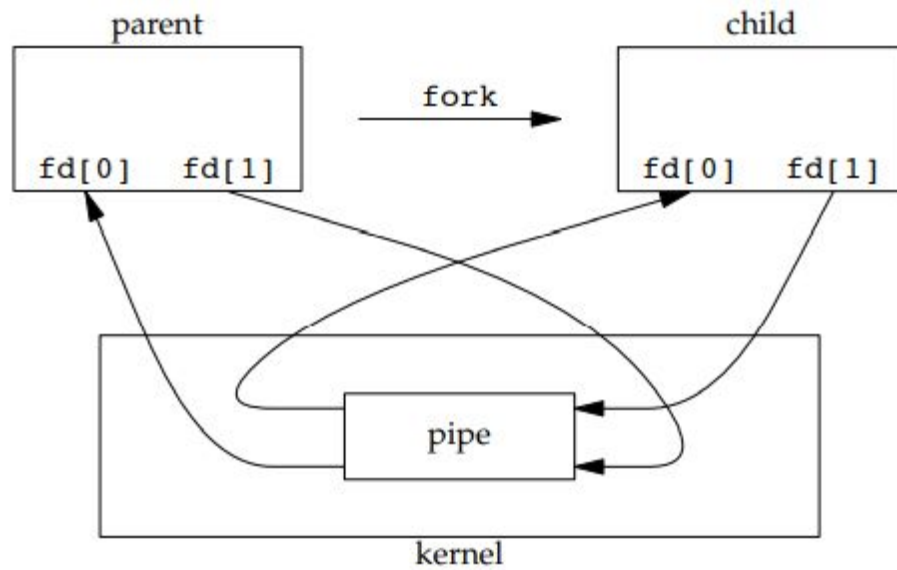
1. The two ends of the pipe connected in a single process



2. The data in the pipe flows through the kernel

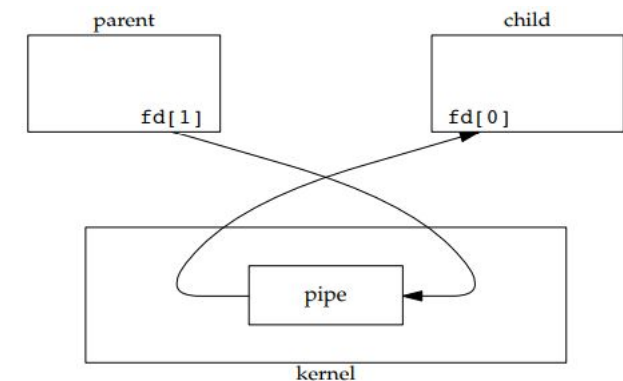
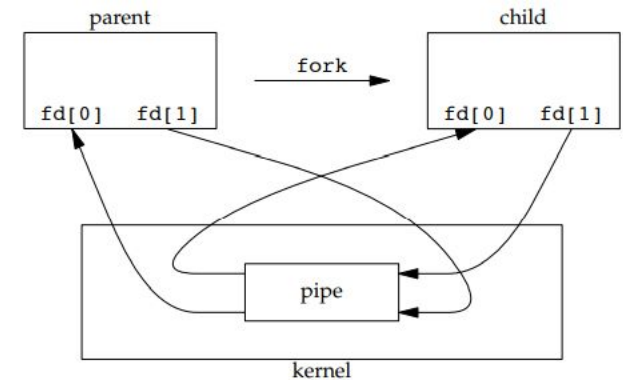






Pipe from parent to child

- Normally, a process that calls pipe then calls fork, creating an IPC channel from the parent to the child or vice versa
- For a pipe from the parent to the child, the parent closes the read end of the pipe (fd[0]) and the child closes the write end (fd[1]).
- For a pipe from the child to the parent, the parent closes fd[1], and the child closes fd[0]



Pipe from parent to child

- A common operation is to create a pipe to another process to either read its output or send it input, the standard I/O library has historically provided the **popen()** and **pclose()** functions
- These two functions handle all the work: creating a pipe, forking a child, closing the unused ends of the pipe, executing a shell to run the command, and waiting for the command to terminate.

```
#include <stdio.h>

FILE *popen(const char *cmdstring, const char *type);

int pclose(FILE *fp);
```

- The function **popen()** does a fork and exec to execute the cmdstring and returns a standard I/O file pointer.
- If type is "r", the file pointer is connected to the standard output of cmdstring.
- If type is "w", the file pointer is connected to the standard input of cmdstring
- The function popen() returns file pointer if OK, NULL on error
- The function **pclose()** closes the standard I/O stream, waits for the command to terminate, and returns the termination status of the shell or -1 on error

- FIFOs are sometimes called **named pipes**.
- Unnamed pipes can be used only between related processes when a common ancestor has created the pipe.
- With FIFOs, unrelated processes can exchange data
- Creating a FIFO is similar to creating a file

```
#include <sys/stat.h>

int mkfifo(const char *path, mode_t mode);

int mkfifoat(int fd, const char *path, mode_t mode);
```

- Functions mkfifo() and mkfifoat() return 0 if OK, -1 on error
- Once a FIFO has been created using mkfifo() or mkfifoat() , it can be opened using open(). Normal file I/O functions (e.g., close, read, write, unlink) all work with FIFOs.
- There are two uses for FIFOs.
 1. FIFOs are used by shell commands to pass data from one shell pipeline to another without creating intermediate temporary files.
 2. FIFOs are used as rendezvous points in client–server applications to pass data between the clients and the servers

- A message queue is a linked list of messages stored within the kernel and identified by a message queue identifier
- A new queue is created or an existing queue opened by **msgget**.

```
#include <sys/msg.h>

int msgget(key_t key, int flag);
```

- Returns message queue ID if OK, -1 on error
- New messages are added to the end of a queue by **msgsnd**.

```
#include <sys/msg.h>

int msgsnd(int msqid, const void *ptr, size_t nbytes, int flag);
```

- Every message has a positive long integer type field, a non-negative length, and the actual data bytes all of which are specified to **msgsnd** when the message is added to a queue.
- Messages are fetched from a queue by **msgrcv**.

```
#include <sys/msg.h>

ssize_t msgrcv(int msqid, void *ptr, size_t nbytes, long type, int flag);
```

- Messages need not be fetched in a first-in, first-out order. Instead, can be fetched based on their type field

- A semaphore is a counter used to provide access to a shared data object for multiple processes.
- To obtain a shared resource, a process needs to do the following:
 1. Test the semaphore that controls the resource.
 2. If the value of the semaphore is positive, the process can use the resource. In this case, the process decrements the semaphore value by 1, indicating that it has used one unit of the resource.
 3. Otherwise, if the value of the semaphore is 0, the process goes to sleep until the semaphore value is greater than 0. When the process wakes up, it returns to step 1.
- When a process is done with a shared resource that is controlled by a semaphore, the semaphore value is incremented by 1.
- If any other processes are asleep, waiting for the semaphore, they are awakened.

- First need to obtain a semaphore ID by calling the `semget` function

```
#include <sys/sem.h>
int semget(key_t key, int nsems, int flag);
```

- Returns: semaphore ID if OK, `-1` on error
- **semctl** function is the catchall for various semaphore operations.

```
#include <sys/sem.h>
int semctl(int semid, int semnum, int cmd, ... /* union semun arg */ );
```

- The function **semop** atomically performs an array of operations on a semaphore set

```
#include <sys/sem.h>
int semop(int semid, struct sembuf semoparray[], size_t nops);
```

- The `semoparray` argument is a pointer to an array of semaphore operations
- Returns 0 if OK, `-1` on error

- Shared memory allows two or more processes to share a given region of memory.
- This is the fastest form of IPC, because the data does not need to be copied between the client and the server
- The challenge in using shared memory is synchronizing access to a given region among multiple processes.
- Semaphores and mutexes are used to synchronize shared memory access
- Function **shmget** is used to obtain a shared memory identifier

```
#include <sys/shm.h>  
  
int shmget(key_t key, size_t size, int flag);
```

- Returns shared memory ID if OK, -1 on error

- Function **shmctl** is the catchall for various shared memory operations

```
#include <sys/shm.h>

int shmctl(int shmid, int cmd, struct shmid_ds *buf);
```

- Returns 0 if OK, -1 on error
- The cmd argument specifies one of the commands to be performed on the segment specified by shmid
- Once a shared memory segment has been created, a process is attached to this memory segment by calling **shmat**

```
#include <sys/shm.h>

void *shmat(int shmid, const void *addr, int flag);
```

- Returns: pointer to shared memory segment if OK, -1 on error

- Function call **shmdt** will detach a shared memory segment from a process

```
#include <sys/shm.h>

int shmdt(const void *addr);
```

- Returns 0 if OK, -1 on error
- A shared memory segment can be removed by calling **shmctl** with a command of IPC_RMID.



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Threads and Concurrency

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Threads and Concurrency

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Overview and Motivation

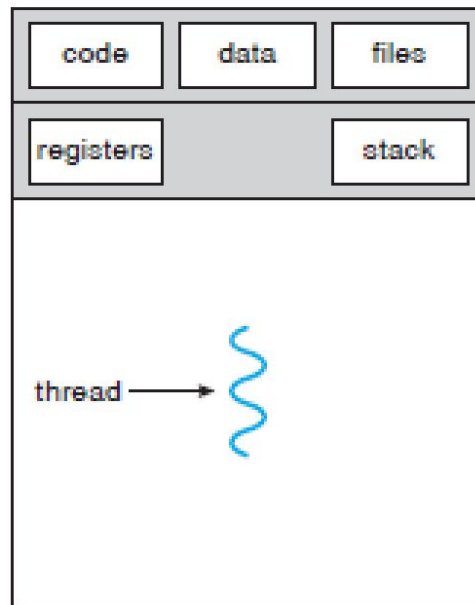
- A Thread is a fundamental unit of CPU utilization that forms the basis of multithreaded computer systems.
- It consists of thread ID, Program counter, a register set and stack
- Shares with other threads of same process its code, data, file descriptors, signals
- Most modern applications are multithreaded
- Threads run within application
- Multiple tasks with the application can be implemented by separate threads.
- Application 1: internet browser.
 - numerous tabs open at a given time
 - Multiple threads of execution are used to load content, display animations, play a video, fetch data from a network and so on.



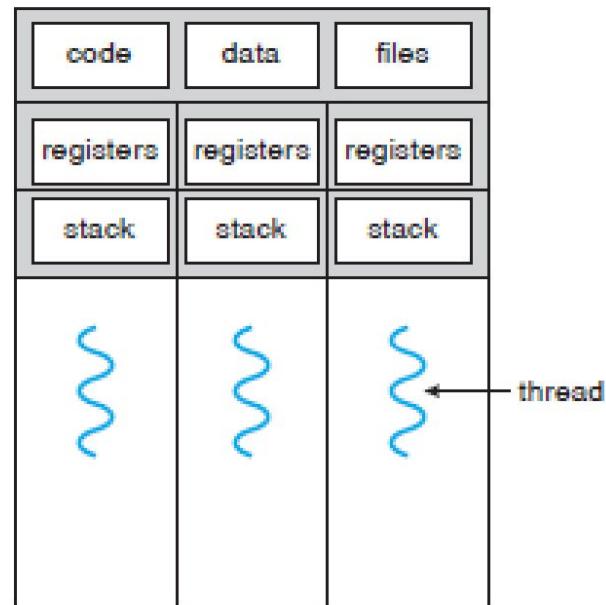
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Single threaded and multithreaded processes

- Application 2: Word processor
 - Thread to accept input
 - A thread for spell checking
 - A thread for grammar checking



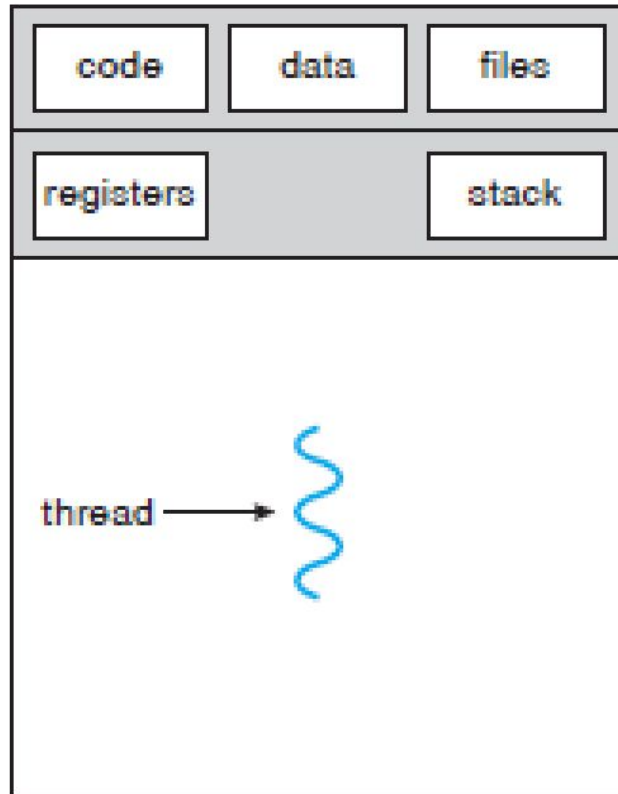
single-threaded process



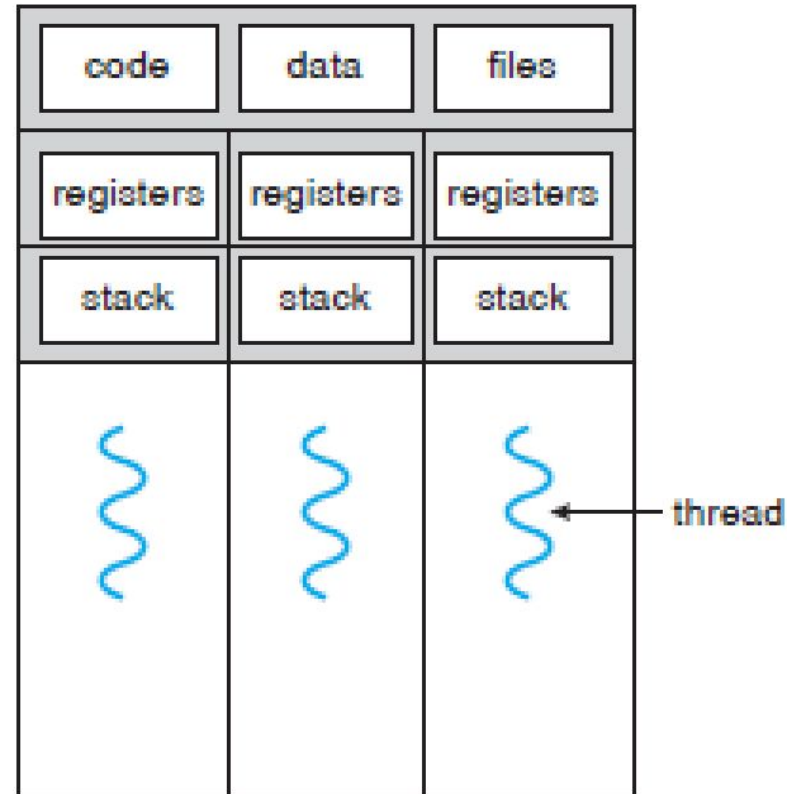
multithreaded process

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Single threaded and multithreaded processes



single-threaded process

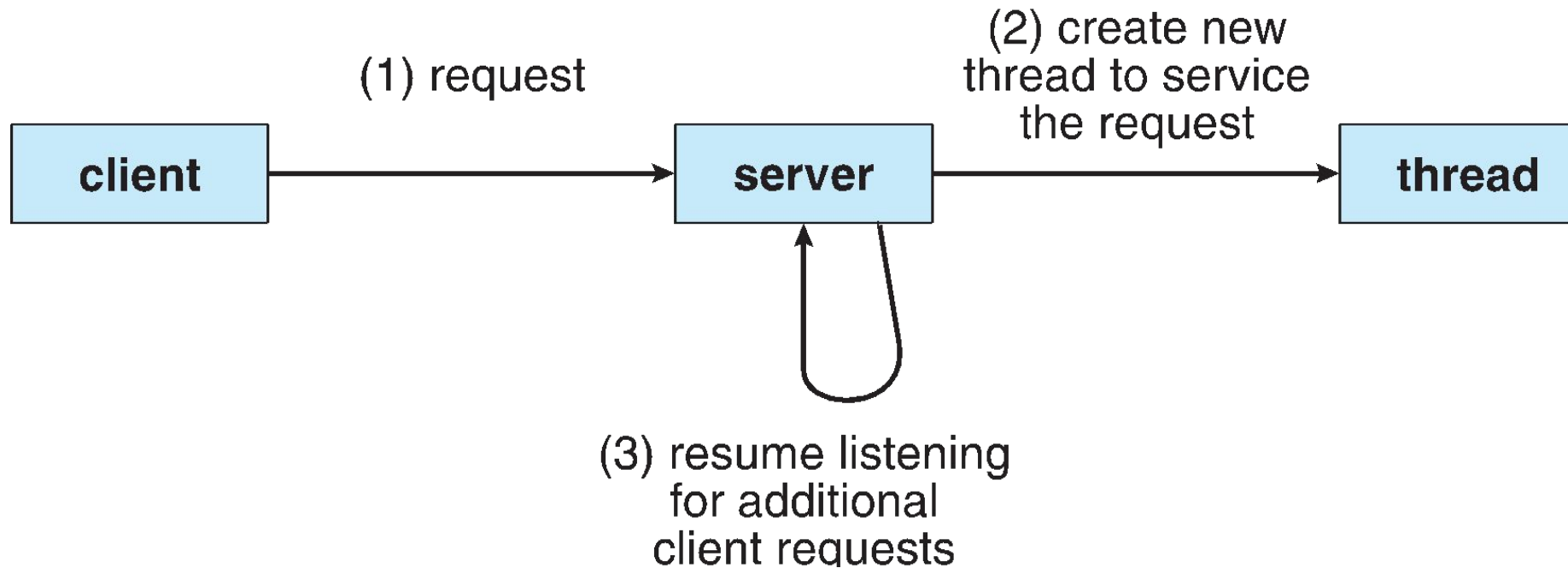


multithreaded process



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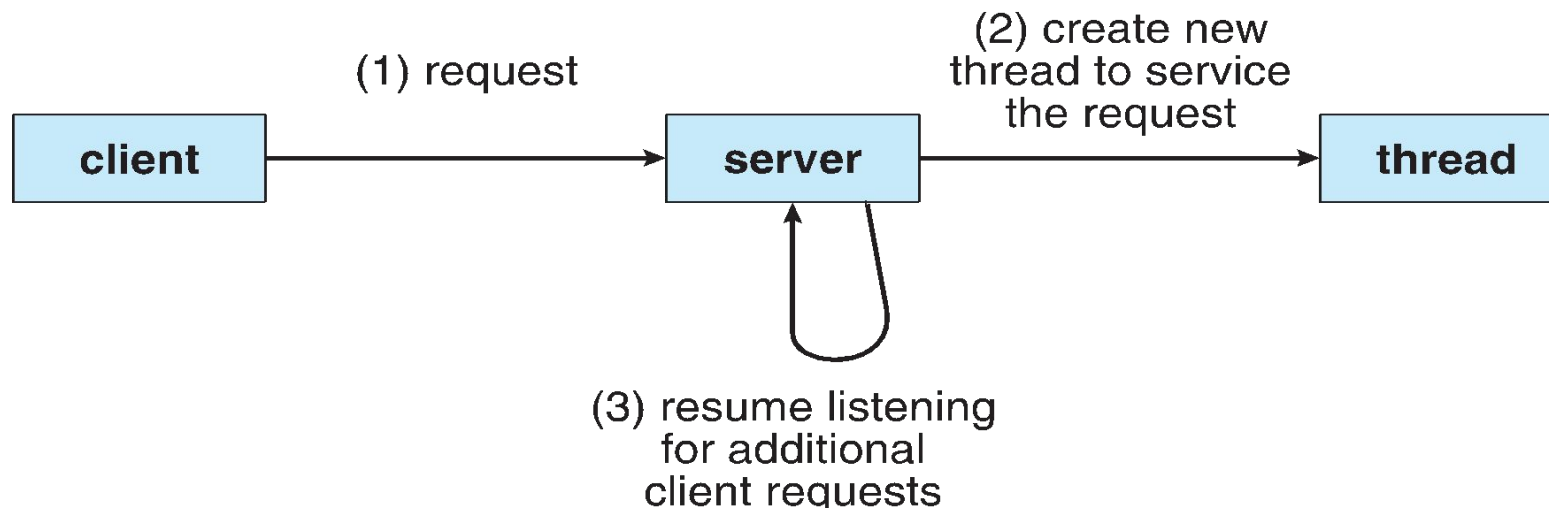
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Multithreaded Server Architecture

- Process creation is heavy-weight while thread creation is light-weight
- Process creation is time consuming, resource intensive
- Threads also play a vital role in remote procedure call (RPC) systems
- Can simplify code, increase efficiency
- Kernels are generally multithreaded



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Benefits

- **Responsiveness**
- **Resource Sharing**
- **Economy**
- **Scalability**



- **Responsiveness** – may allow continued execution if part of process is blocked or performing lengthy operations.
 - It is useful in designing user interfaces.
 - A multithreaded Web browser could allow user interaction in one thread while an image was being loaded in another thread.
 - Example: click on the button
- **Resource Sharing** – threads share resources of process, easier than shared memory or message passing.
 - Programmer needs to specify the techniques for sharing
 - But threads share the memory and other resources
 - Sharing of resources helps in having many threads within the same address space

- **Economy** – cheaper than process creation, thread switching lower overhead than context switching.
 - In Solaris, for example, creating a process is about thirty times slower than creating a thread, and context switching is about five times slower.
- **Scalability** – process can take advantage of multiprocessor architectures.
 - Threads can run on multiple cores parallelly

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Processes vs Threads

Process

- Will by default not share memory
- Most file descriptors not shared
- Don't share filesystem context
- Don't share signal handling

Thread

- Will by default share memory
- Will share file descriptors
- Will share filesystem context
- Will share signal handling



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Attributes shared by Threads

- ▣ process ID and parent process ID; process group ID and session ID;
- ▣ controlling terminal;
- ▣ process credentials (user and group IDs); open file descriptors;
- ▣ record locks created using `fcntl()`; signal dispositions;
- ▣ file system–related information: `umask`, `cwd` and root directory;
- ▣ resource limits; CPU time consumed (as returned by `times()`);
- ▣ resources consumed (as returned by `getrusage()`); nice value (set by `setpriority()` and `nice()`).



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Attributes specific to Threads



- ▣ thread ID ;signal mask;
- ▣ Thread-specific data ;
- ▣ the errno variable;
- ▣ floating-point environment (see fenv(3));
- ▣ stack (local variables and function call linkage information i.e CPU registers saved in the called function's stack frame when one function calls another function and restored for the calling function when the called function returns)
- ▣ and a few more

- A thread consists of the following information necessary to represent an execution context within a process.
- thread ID that identifies the thread within a process
- Every thread has a thread ID
- set of register values
- stack
- scheduling priority and policy
- signal mask
- Errno variable
- Thread-specific/thread-private data (each thread to access its own separate copy of the data, without worrying about synchronizing access with other threads)

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Thread Concepts (Cont.)

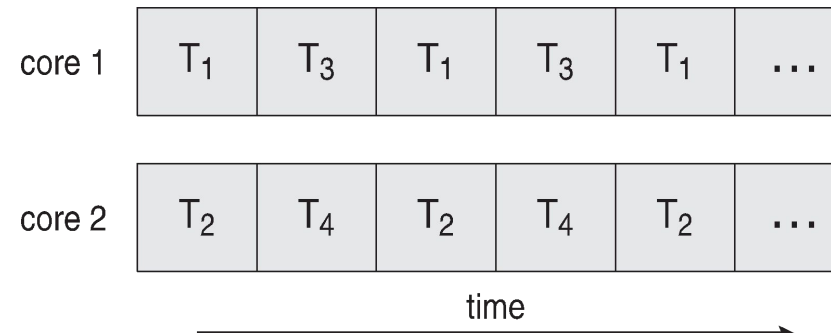
- When a thread is created, there is no guarantee which will run first: the newly created thread or the calling thread.
- The newly created thread has access to the process address space and inherits the calling thread's floating-point environment and signal mask
- The set of pending signals for the thread is cleared.
- The pthread functions usually return an error code when they fail. They don't set errno like the other POSIX functions.



- **Concurrent execution on single-core system:**



- **Parallelism on a multi-core system:**



- **Multicore** or **multiprocessor** systems putting pressure on programmers, challenges include:
 - **Dividing activities**
 - **Balance**
 - **Data splitting**
 - **Data dependency**
 - **Testing and debugging**
- **Parallelism** implies a system can perform more than one task simultaneously
- **Concurrency** supports more than one task making progress
 - Single processor / core, scheduler providing concurrency

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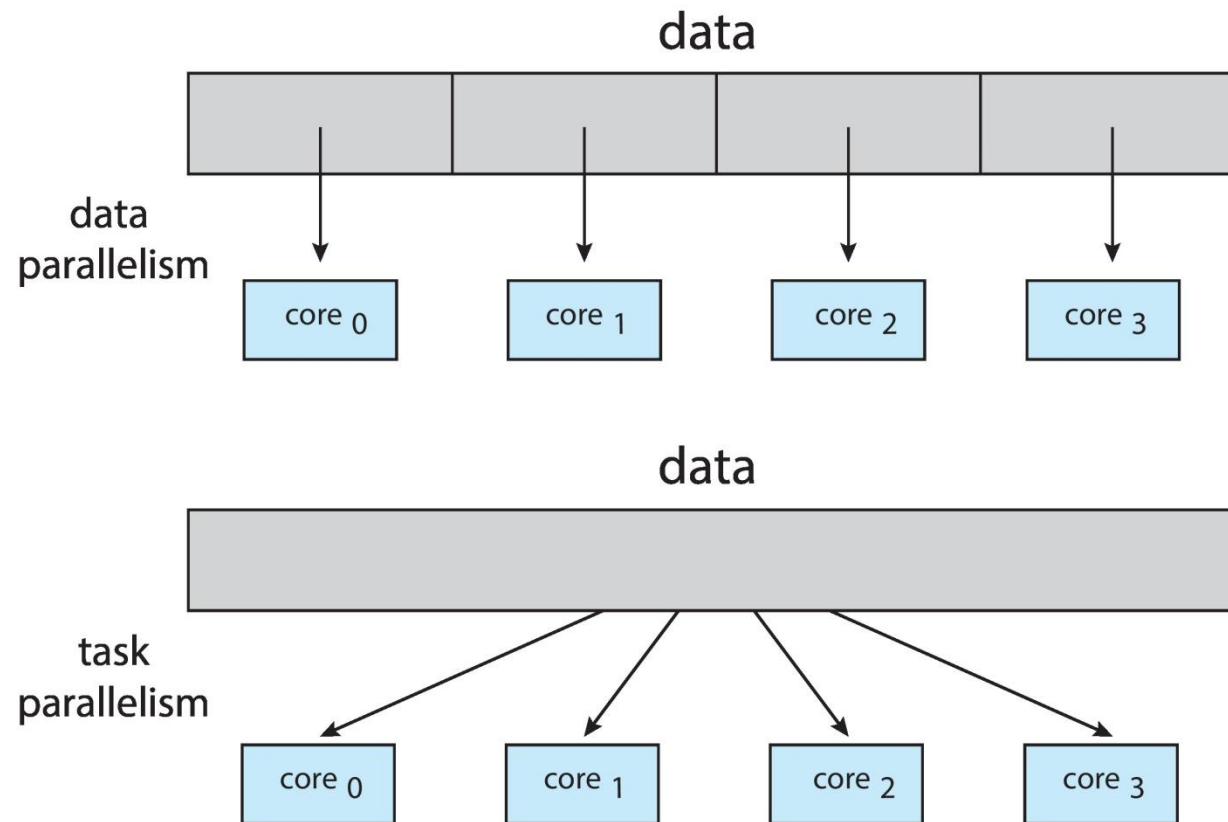
Multicore Programming (Cont.)

- Types of parallelism
 - **Data parallelism** – distributes subsets of the same data across multiple cores, same operation on each
 - **Task parallelism** – distributing threads across cores, each thread performing unique operation
- As # of threads grows, so does architectural support for threading
 - CPUs have cores as well as ***hardware threads***
 - Consider Oracle SPARC T4 with 8 cores, and 8 hardware threads per core



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Data and Task Parallelism



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Amdahl's Law



- Identifies performance gains from adding additional cores to an application that has both serial and parallel components
- S is portion of an application that needs to be done in serial
- N processing cores

$$\text{speedup} \leq \frac{1}{S + \frac{(1-S)}{N}}$$

- That is, if application is 75% parallel / 25% serial, moving from 1 to 2 cores results in speedup of 1.6 times
- As N approaches infinity, speedup approaches $1 / S$
Serial portion of an application has disproportionate effect on performance gained by adding additional cores
- But does the law take into account contemporary multicore systems?

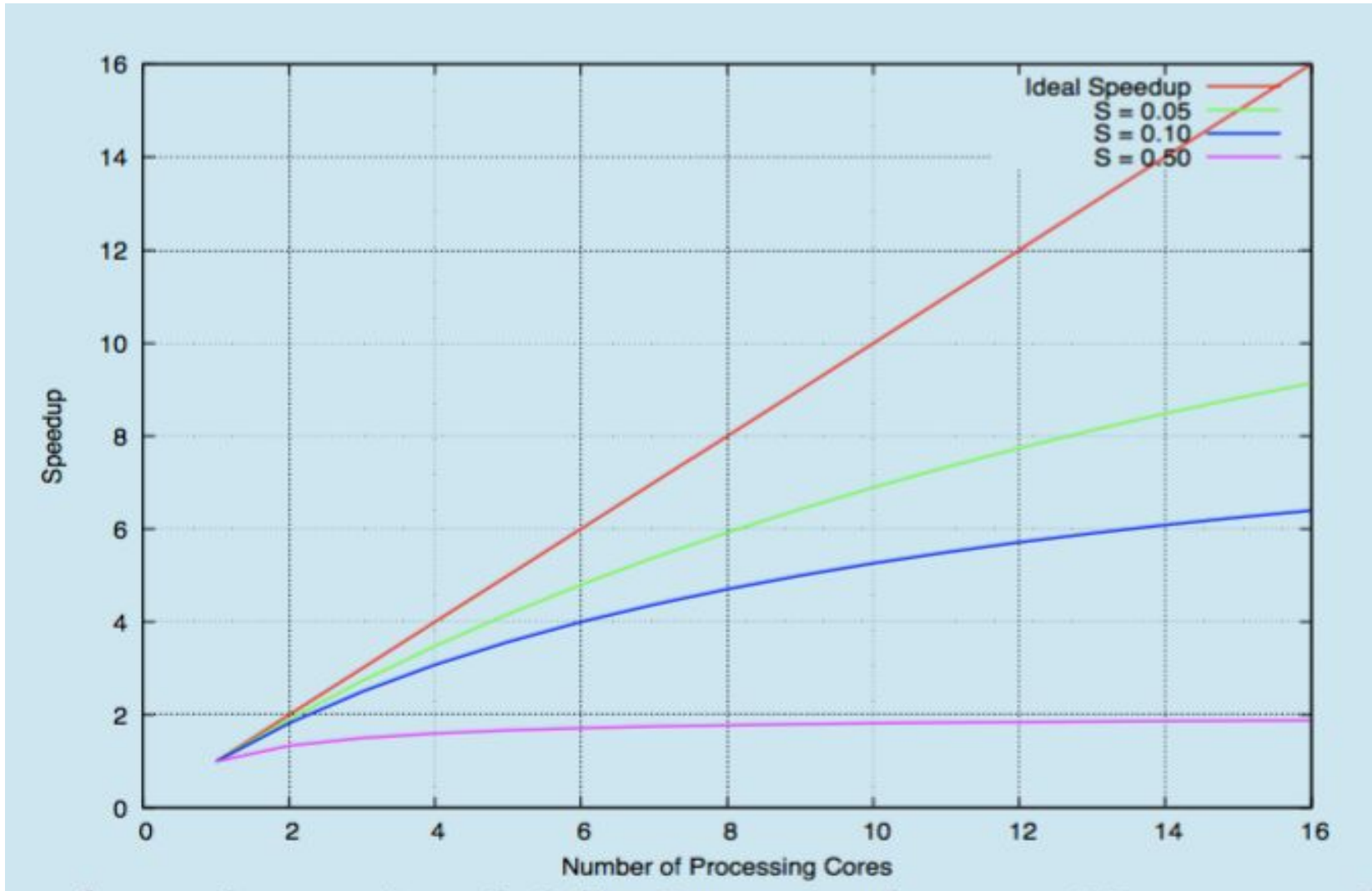
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Amdahl's Law



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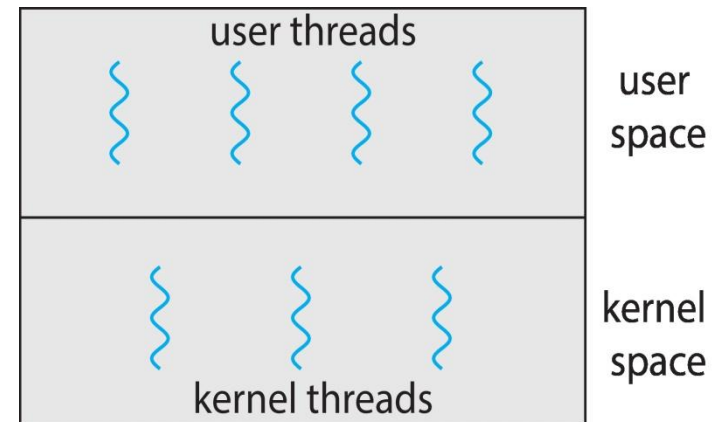
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User Threads and Kernel Threads

- **User threads** - management done by user-level threads library
- Three primary thread libraries:
 - POSIX **Pthreads**
 - Windows threads
 - Java threads
- **Kernel threads** - Supported by the Kernel
- Examples – virtually all general purpose operating systems, including:
 - Windows
 - Solaris
 - Linux
 - Tru64 UNIX
 - Mac OS X





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OPERATING SYSTEMS

UE24CS242B

Multithreading Models, Thread Creation and Scheduling

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OPERATING SYSTEMS

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Multithreading Models, Thread Creation and Scheduling

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Multithreading Models

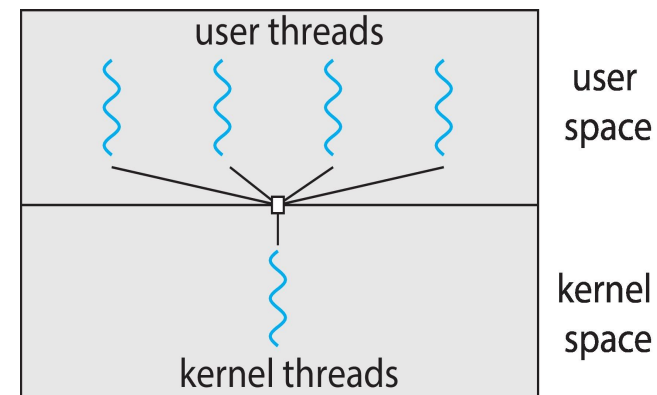
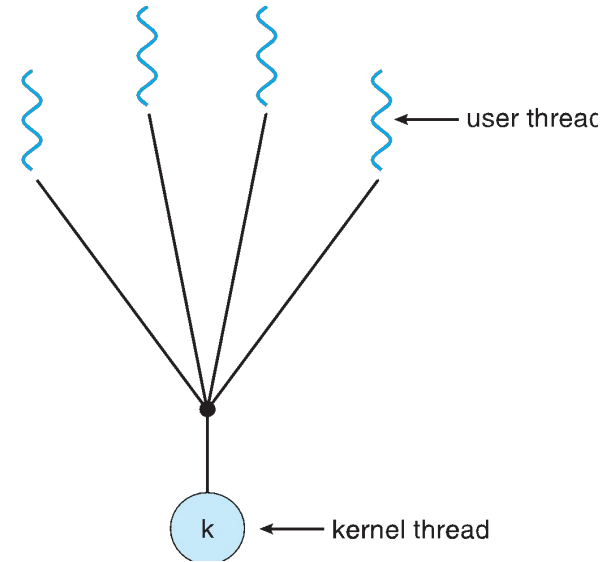
- Many-to-One
- One-to-One
- Many-to-Many



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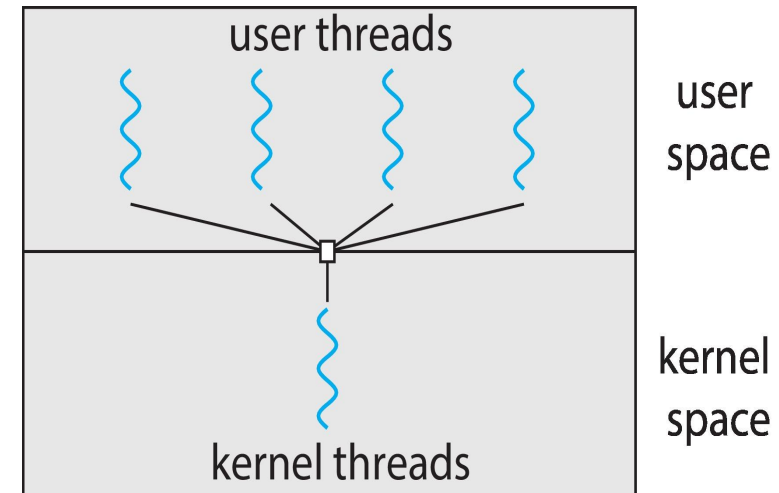
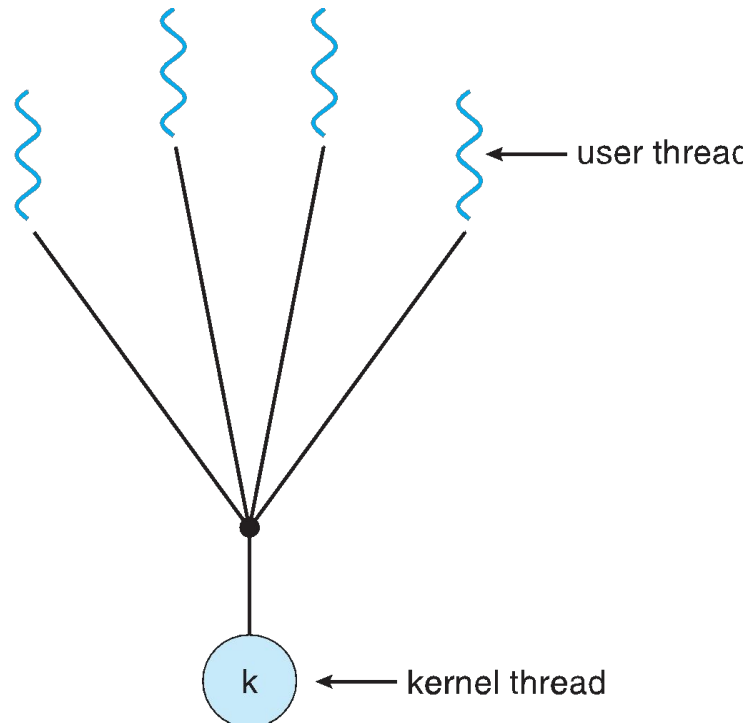
Many-to-One

- Many user-level threads mapped to single kernel thread
- One thread blocking causes all to block
- Multiple threads may not run in parallel on multicore system because only one may be in kernel at a time
- Few systems currently use this model
- Examples:
 - **Solaris Green Threads**
 - **GNU Portable Threads**



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Many-to-One



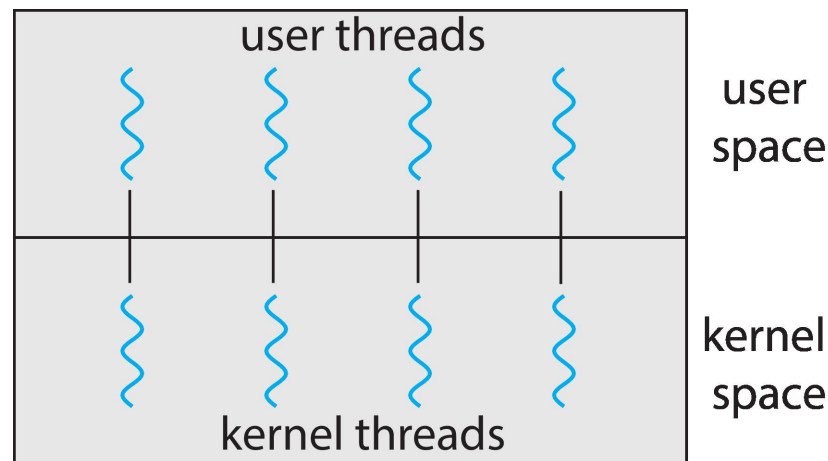
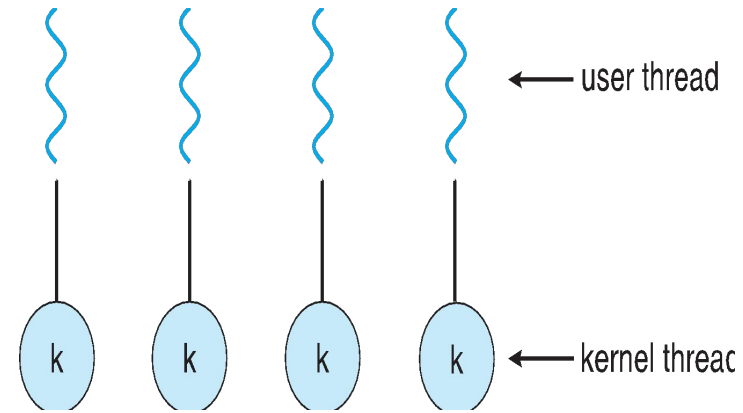
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One-to-One

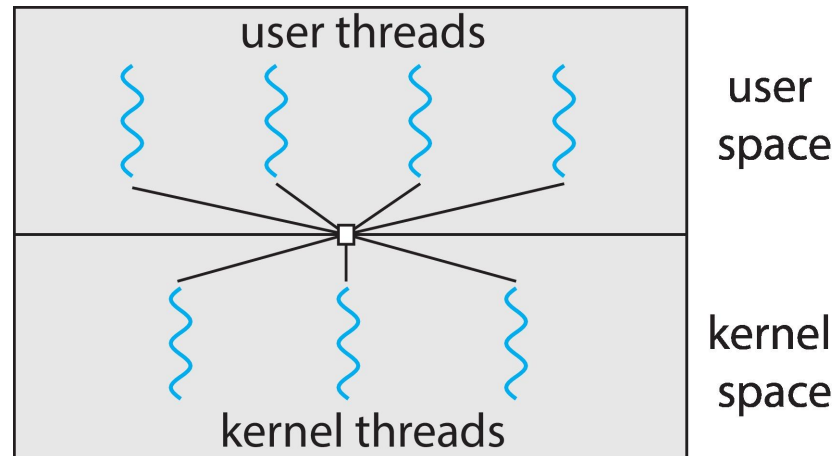
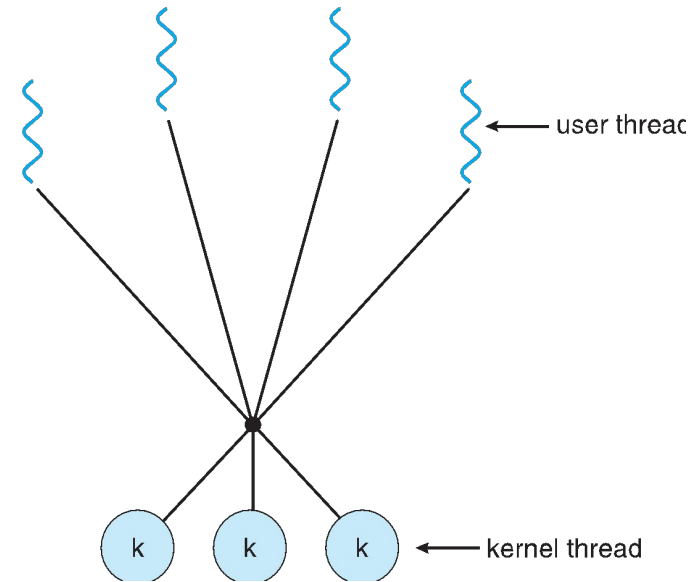
- Each user-level thread maps to kernel thread
- Creating a user-level thread creates a kernel thread
- More concurrency than many-to-one
- Number of threads per process sometimes restricted due to overhead
- Examples
 - Windows
 - Linux
 - Solaris 9 and later



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Many-to-Many Model

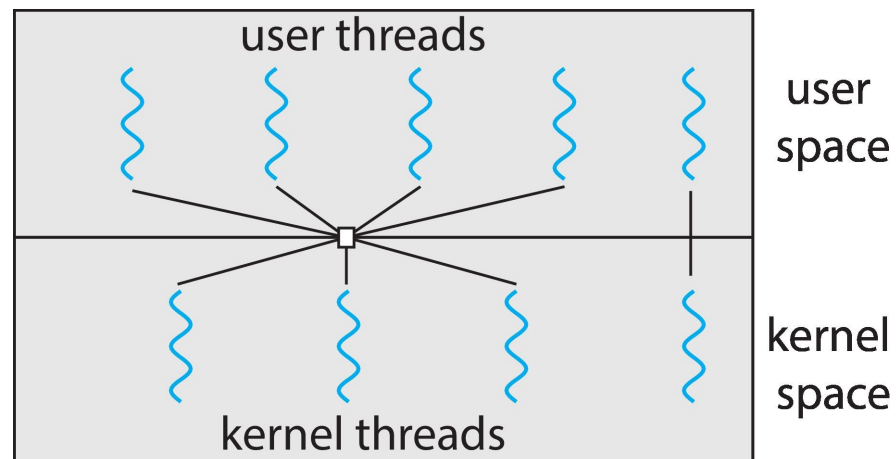
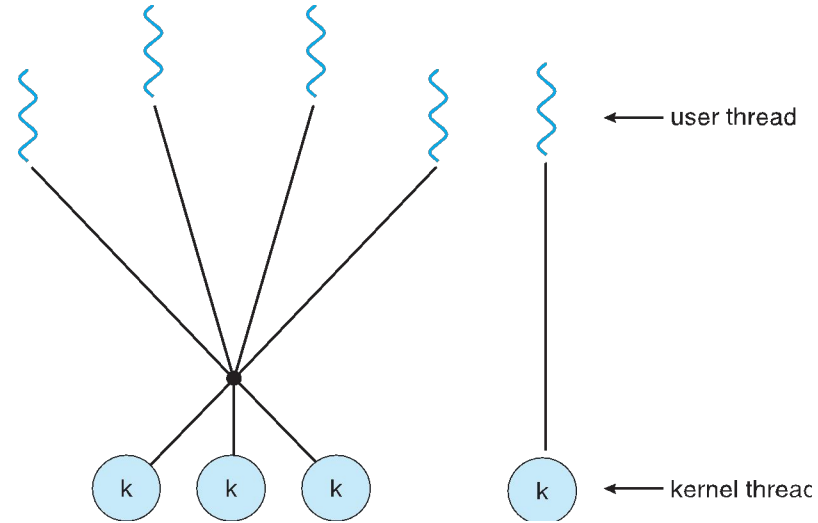
- Allows many user level threads to be mapped to many kernel threads
- Allows the operating system to create a sufficient number of kernel threads
- Solaris prior to version 9
- Windows with the *ThreadFiber* package



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Two-level Model

- Similar to M:M, except that it allows a user thread to be **bound** to kernel thread
- Examples
 - IRIX
 - HP-UX
 - Tru64 UNIX
 - Solaris 8 and earlier



- May be provided either as user-level or kernel-level
- A POSIX standard (IEEE 1003.1c) API for thread creation and synchronization
- ***Specification***, not ***implementation***
- API specifies behavior of the thread library, implementation is up to development of the library
- Common in UNIX operating systems (Solaris, Linux, Mac OS X)

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Pthreads Example

```
#include <pthread.h>
#include <stdio.h>

int sum; /* this data is shared by the thread(s) */
void *runner(void *param); /* threads call this function */

int main(int argc, char *argv[])
{
    pthread_t tid; /* the thread identifier */
    pthread_attr_t attr; /* set of thread attributes */

    if (argc != 2) {
        fprintf(stderr, "usage: a.out <integer value>\n");
        return -1;
    }
    if (atoi(argv[1]) < 0) {
        fprintf(stderr, "%d must be >= 0\n", atoi(argv[1]));
        return -1;
    }
}
```



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Pthreads Example

```
/* get the default attributes */
pthread_attr_init(&attr);
/* create the thread */
pthread_create(&tid,&attr,runner,argv[1]);
/* wait for the thread to exit */
pthread_join(tid,NULL);

printf("sum = %d\n",sum);
}

/* The thread will begin control in this function */
void *runner(void *param)
{
    int i, upper = atoi(param);
    sum = 0;

    for (i = 1; i <= upper; i++)
        sum += i;

    pthread_exit(0);
}
```



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Pthreads Code for Joining 10 Threads

```
#define NUM_THREADS 10

/* an array of threads to be joined upon */
pthread_t workers[NUM_THREADS];

for (int i = 0; i < NUM_THREADS; i++)
    pthread_join(workers[i], NULL);
```



- Distinction between user-level and kernel-level threads
- When threads supported, threads scheduled, not processes
- Many-to-one and many-to-many models, thread library schedules user-level threads to run on LWP.
- Scheduling of user level threads (ULT) to kernel level threads (KLT) via leight-weight process (LWP) by the application developer.
 - **Leightweight Process (LWP)** : Light-weight process are threads in the user space that acts as an interface for the ULT to access the physical CPU resources
- Scheduling of kernel level threads by the system scheduler to perform different unique OS functions.

- Many-to-one and many-to-many models, thread library schedules user-level threads to run on LWP
 - Known as **process-contention scope (PCS)** since scheduling competition is within the process
 - Typically done via priority set by programmer
- Kernel thread scheduled onto available CPU is **system-contention scope (SCS)** – competition among all threads in system

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Pthread Scheduling

- API allows specifying either PCS or SCS during thread creation
 - PTHREAD_SCOPE_PROCESS schedules threads using PCS scheduling
 - PTHREAD_SCOPE_SYSTEM schedules threads using SCS scheduling
- Can be limited by OS – Linux and Mac OS X only allow PTHREAD_SCOPE_SYSTEM



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Pthread Scheduling API

```
#include <pthread.h>
#include <stdio.h>
#define NUM_THREADS 5
int main(int argc, char *argv[]) {
    int i, scope;
    pthread_t tid[NUM_THREADS];
    pthread_attr_t attr;
    /* get the default attributes */
    pthread_attr_init(&attr);
    /* first inquire on the current scope */
    if (pthread_attr_getscope(&attr, &scope) != 0)
        fprintf(stderr, "Unable to get scheduling scope\n");
    else {
        if (scope == PTHREAD_SCOPE_PROCESS)
            printf("PTHREAD_SCOPE_PROCESS");
        else if (scope == PTHREAD_SCOPE_SYSTEM)
            printf("PTHREAD_SCOPE_SYSTEM");
        else
            fprintf(stderr, "Illegal scope value.\n");
    }
}
```



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Pthread Scheduling API

```
/* set the scheduling algorithm to PCS or SCS */
pthread_attr_setscope(&attr, PTHREAD_SCOPE_SYSTEM);
/* create the threads */
for (i = 0; i < NUM_THREADS; i++)
    pthread_create(&tid[i], &attr, runner, NULL);
/* now join on each thread */
for (i = 0; i < NUM_THREADS; i++)
    pthread_join(tid[i], NULL);
}
/* Each thread will begin control in this function */
void *runner(void *param)
{
    /* do some work ... */
    pthread_exit(0);
}
```





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Threads and Concurrency

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Pthreads and Windows Threads

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- Provides the programmer with an API for creating and managing threads.
- Two primary ways of implementing a thread library.
 - Provide a library entirely in user space with no kernel support.
 - All code and data structures for the library exist in user space.
 - Implement a kernel-level library
 - Code and data structures for the library exist in kernel space.
 - Invoking a function directly by the operating system.

- Three main thread libraries are in use today:
 - (1) POSIX Pthreads,
 - (2) Win32,
 - (3) Java.
- Pthreads, the threads extension of the POSIX standard, may be provided as either a user- or kernel-level library.
- The Win32 thread library is a kernel-level library available on Windows systems.
- The Java thread API allows threads to be created and managed directly in Java programs.

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Thread Libraries: Pthreads

- May be provided either as user-level or kernel-level
- A POSIX standard (IEEE 1003.1c) API for thread creation and synchronization
- ***Specification***, not ***implementation***
- API specifies behavior of the thread library, implementation is up to development of the library
- Common in UNIX operating systems (Solaris, Linux, Mac OS X)



```
#include <pthread.h>
#include <stdio.h>

int sum; /* this data is shared by the thread(s) */
void *runner(void *param); /* threads call this function */

int main(int argc, char *argv[])
{
    pthread_t tid; /* the thread identifier */
    pthread_attr_t attr; /* set of thread attributes */

    if (argc != 2) {
        fprintf(stderr, "usage: a.out <integer value>\n");
        return -1;
    }
    if (atoi(argv[1]) < 0) {
        fprintf(stderr, "%d must be >= 0\n", atoi(argv[1]));
        return -1;
    }
}
```

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Pthreads Example

```
/* get the default attributes */
pthread_attr_init(&attr);
/* create the thread */
pthread_create(&tid,&attr,runner,argv[1]);
/* wait for the thread to exit */
pthread_join(tid,NULL);

printf("sum = %d\n",sum);
}

/* The thread will begin control in this function */
void *runner(void *param)
{
    int i, upper = atoi(param);
    sum = 0;

    for (i = 1; i <= upper; i++)
        sum += i;

    pthread_exit(0);
}
```



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Pthreads Code for Joining 10 Threads

```
#define NUM_THREADS 10

/* an array of threads to be joined upon */
pthread_t workers[NUM_THREADS];

for (int i = 0; i < NUM_THREADS; i++)
    pthread_join(workers[i], NULL);
```



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Windows threads

- ☐ Windows implements the Win32 API as its primary API.
- ☐ A Windows application runs as a separate process, and each process may contain one or more threads.
- ☐ Windows uses the one-to-one mapping.
- ☐ Windows also provides support for a **fiber** library, which provides the functionality of the many-to-many model



- ❑ The general components of a thread include:
 - A thread ID uniquely identifying the thread
 - A register set representing the status of the processor.
 - A user stack, employed when the thread is running in user mode, and a kernel stack, employed when the thread is running in kernel mode
 - A private storage area various run-time libraries and dynamic link libraries (DLLs)

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Windows Threads

- Threads are created in the Win32 API using the `CreateThread()` function the Win32 API.
- Using the `WaitForSingleObject()` function, which causes the creating thread to block until the child thread has exited



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Windows Multithreaded C Program

```
#include <windows.h>
#include <stdio.h>
DWORD Sum; /* data is shared by the thread(s) */

/* the thread runs in this separate function */
DWORD WINAPI Summation(LPVOID Param)
{
    DWORD Upper = *(DWORD*)Param;
    for (DWORD i = 0; i <= Upper; i++)
        Sum += i;
    return 0;
}

int main(int argc, char *argv[])
{
    DWORD ThreadId;
    HANDLE ThreadHandle;
    int Param;

    if (argc != 2) {
        fprintf(stderr, "An integer parameter is required\n");
        return -1;
    }
    Param = atoi(argv[1]);
    if (Param < 0) {
        fprintf(stderr, "An integer >= 0 is required\n");
        return -1;
    }
}
```



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Windows Multithreaded C Program (Cont.)

```
/* create the thread */
ThreadHandle = CreateThread(
    NULL, /* default security attributes */
    0, /* default stack size */
    Summation, /* thread function */
    &Param, /* parameter to thread function */
    0, /* default creation flags */
    &ThreadId); /* returns the thread identifier */

if (ThreadHandle != NULL) {
    /* now wait for the thread to finish */
    WaitForSingleObject(ThreadHandle, INFINITE);

    /* close the thread handle */
    CloseHandle(ThreadHandle);

    printf("sum = %d\n", Sum);
}
}
```





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Mutual Exclusion and Synchronization

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Software Approaches

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- Processes can execute concurrently
 - May be interrupted at any time, partially completing execution
- Concurrent access to shared data may result in data inconsistency
- Maintaining data consistency requires mechanisms to ensure the orderly execution of cooperating processes
- Illustration of the problem:

Suppose that we wanted to provide a solution to the consumer-producer problem that fills ***all*** the buffers. We can do so by having an integer **counter** that keeps track of the number of full buffers. Initially, **counter** is set to 0. It is incremented by the producer after it produces a new buffer and is decremented by the consumer after it consumes a buffer.


```
while (true) {  
    /* produce an item in next_produced */  
  
    while (counter == BUFFER_SIZE) ;  
        /* do nothing */  
    buffer[in] = next_produced;  
    in = (in + 1) % BUFFER_SIZE; /* Variable in points to the next free position in the buffer */  
    counter++;  
}
```

```
while (true) {  
    while (counter == 0)  
        ; /* do nothing */  
    next_consumed = buffer[out];  
    out = (out + 1) % BUFFER_SIZE; /* Variable out points to the first full position in the buffer */  
    counter--;  
    /* consume the item in next consumed */  
}
```

- **counter++** could be implemented as

```
register1 = counter  
register1 = register1 + 1  
counter = register1
```

- **counter--** could be implemented as

```
register2 = counter  
register2 = register2 - 1  
counter = register2
```

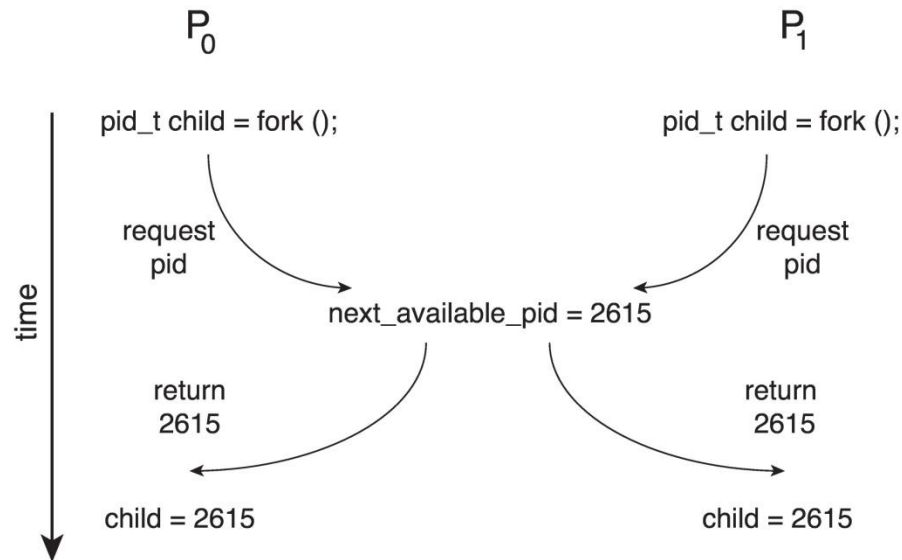
- Consider this execution interleaving with “count = 5” initially:

S0: producer execute	register1 = counter	{register1 = 5}
S1: producer execute	register1 = register1 + 1	{register1 = 6}
S2: consumer execute	register2 = counter	{register2 = 5}
S3: consumer execute	register2 = register2 - 1	{register2 = 4}
S4: producer execute	counter = register1	{counter = 6 }
S5: consumer execute	counter = register2	{counter = 4}

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Race Condition (Another Example)

- Processes P_0 and P_1 are creating child processes using the `fork()` system call
- Race condition on kernel variable `next_available_pid` which represents the next available process identifier (pid)



- Unless there is a mechanism to prevent P_0 and P_1 from accessing the variable `next_available_pid` the same pid could be assigned to two different processes!

- Consider system of n processes $\{p_0, p_1, \dots, p_{n-1}\}$
- Each process has **critical section** segment of code
 - Process may be changing common variables, updating table, writing file, etc
 - When one process in critical section, no other may be in its critical section
- **Critical section problem** is to design protocol to solve this
- Each process must ask permission to enter critical section in **entry section**, may follow critical section with **exit section**, then **remainder section**

```
do {  
    entry section  
    critical section  
    exit section  
    remainder section  
} while (true);
```

- General structure of process P_i

```
do {  
    entry section  
    critical section  
    exit section  
    remainder section  
} while (true);
```

1. **Mutual Exclusion** - If process P_i is executing in its critical section, then no other processes can be executing in their critical sections
2. **Progress** - If no process is executing in its critical section and there exist some processes that wish to enter their critical section, then the selection of the processes that will enter the critical section next cannot be postponed indefinitely
3. **Bounded Waiting** - A bound must exist on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted
 - Assume that each process executes at a nonzero speed
 - No assumption concerning **relative speed** of the n processes

Two approaches depending on if kernel is preemptive or non- preemptive

- **Preemptive** – allows preemption of process when running in kernel mode.
 - Preemptive kernels are difficult to design on SMP architectures
- **Non-preemptive** – runs until exits kernel mode, blocks, or voluntarily yields CPU
 - Essentially free of race conditions in kernel mode
 - It is free from race conditions on kernel data structures
- Preemptive kernel may be more responsive. Suitable for real-time systems.

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Peterson's Solution

- Software-based solution to the Critical Section problem.
- Good algorithmic description of solving the problem
- Assume that the **load** and **store** machine-language instructions are atomic; that is, cannot be interrupted
- **Peterson's Solution** restricted to two process.
- P_i and P_j are two process, $j=1-i$
- Peterson solution requires two processes share two data items:
 - `int turn;`
 - `Boolean flag[2]`
- The variable **turn** indicates whose turn it is to enter the critical section
- The **flag** array is used to indicate if a process is ready to enter the critical section. **flag[i] = true** implies that process P_i



```
while (true) {  
    flag[i] = true;  
    turn = j;  
    while (flag[j] && turn == j);  
    /* critical section */  
    flag[i] = false;  
    /* remainder section */  
}
```

The structure of process P_i in Peterson's solution

Structure of process P_i in Peterson's solution

```
do {  
    flag [ i ] = true ;  
    turn = j ;  
    while ( flag [ j ] && turn == [ j ] ) ;  
  
    critical section  
  
    flag [ i ] = false ;  
  
    remainder section  
  
} while (TRUE) ;
```

Structure of process P_j in Peterson's solution

```
do {  
    flag [ j ] = true ;  
    turn = i ;  
    while ( flag [ i ] && turn == [ i ] ) ;  
  
    critical section  
  
    flag [ j ] = false ;  
  
    remainder section  
  
} while (TRUE) ;
```

- To prove solution is correct, we need show
- Mutual exclusion is preserved

P_i enters critical section only if:

turn = i

and **turn** cannot be both 0 and 1 at the same time

- What about the Progress requirement?
- What about the Bounded-waiting requirement?

- Provable that the three CS requirement are met:
 1. Mutual exclusion is preserved
 P_i enters CS only if:
either **flag[j] = false** or **turn = i**
 2. Progress requirement is satisfied
 3. Bounded-waiting requirement is met



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Mutex Locks and Semaphores

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Mutex Locks and Semaphores

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Mutex Locks

- ❑ Previous solutions i.e hardware and software solutions are complicated and generally inaccessible to application programmers
- ❑ OS designers build software tools to solve critical section problem
- ❑ Simplest is mutex lock
- ❑ Protect a critical section by first **acquire()** a lock then **release()** the lock
 - ❑ Boolean variable indicating if lock is available or not
- ❑ Calls to **acquire()** and **release()** must be atomic
 - ❑ Usually implemented via hardware atomic instructions
- ❑ But this solution requires **busy waiting**
- ❑ This lock therefore called a **spinlock**
- ❑ It is problem in real time systems
- ❑ Busy waiting wastes CPU cycles



Advantages of spinlocks

- ❖ no context switch is required when a process must wait on a lock.
- ❖ context switch may take considerable time.
- ❖ When locks are expected to be held for short times, spinlocks are useful
- ❖ These are employed on multiprocessor systems where one thread can “spin” on one processor while another thread performs its critical section on another processor.

```
while (TRUE) {  
    acquire lock  
    critical section  
    release lock  
    remainder section  
}
```

- Implementation of acquire and release

1. While (LOCK==1) ;

2. LOCK = 1

3. Critical Section

4. LOCK = 0

```
while (TRUE) {  
    acquire lock  
    critical section  
    release lock  
    remainder section  
}
```

- Implementation of acquire and release
- **acquire() {
 while (!available)
 ; /* busy wait */
 available = false;
}**
- **release() {
 available = true;
}**

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acquire() and release()

- Solution to a critical section problem using mutex
- **do {**
 - acquire lock*
 - critical section
 - release lock*
 - remainder section
- } while (true);**



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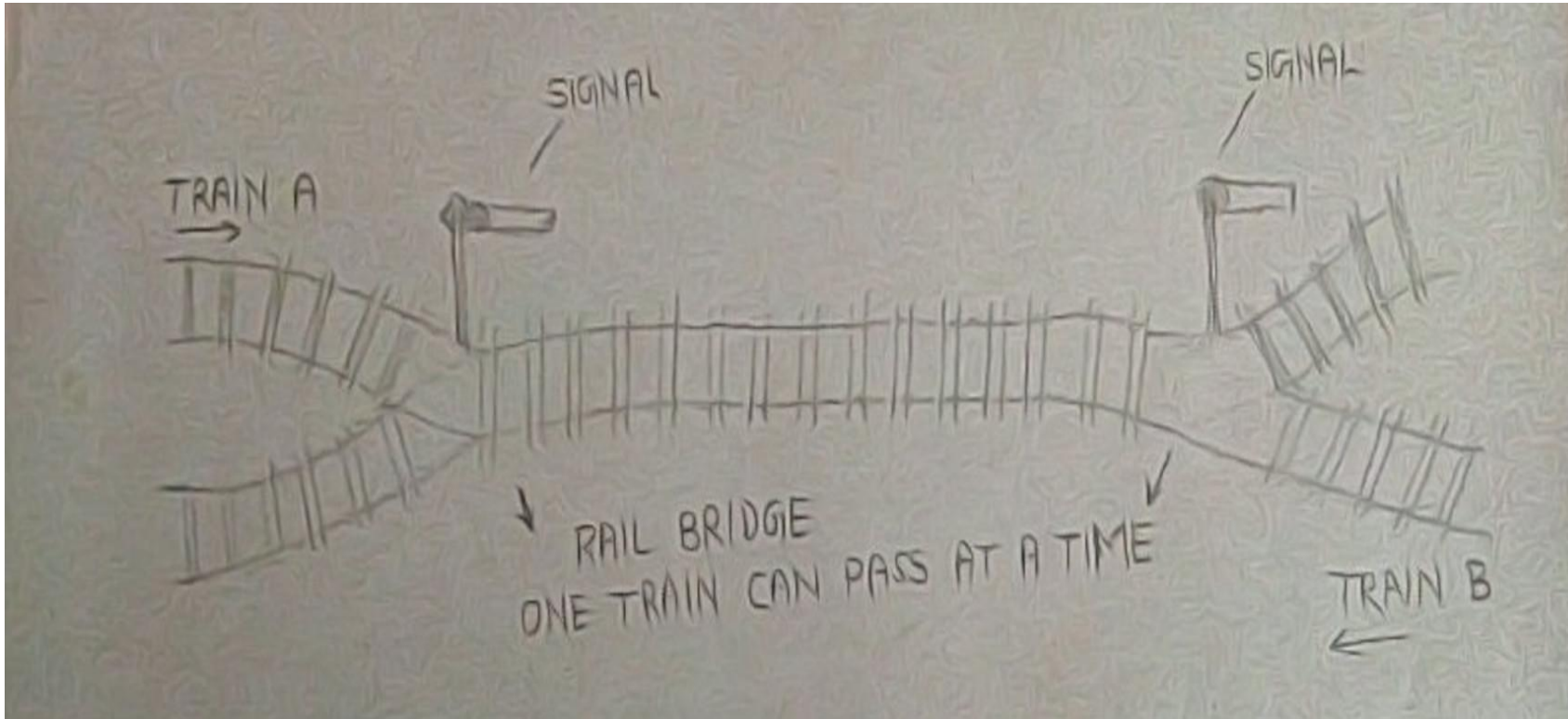
Scenario 1

- Consider a library of an university with 10 rooms
- At a time one room can be used by only one student by informing the front desk for reading.
- Once he completes reading, he has to inform the front desk.
- Person at front desk knows how many rooms are available for use and how many are occupied, how many of them are waiting.
- Once the room is vacant, who will get the chance to occupy the room?



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Scenario 2



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Semaphore

- Synchronization tool that provides more sophisticated ways (than Mutex locks) for process to synchronize their activities.
- Semaphore **S** – integer variable
- Can only be accessed via two indivisible (atomic) operations
 - **wait()** and **signal()**
 - 4 Originally called **P()** and **V()**
- Definition of the **wait()** operation

```
wait(S) {  
    while (S <= 0)  
        ; // busy wait  
    S--;} 
```
- Definition of the **signal()** operation

```
signal(S) {  
    S++;  
}
```



- **Counting semaphore** – integer value can range over an unrestricted domain
- **Binary semaphore** – integer value can range only between 0 and 1
 - Same as a **mutex lock**
- Can solve various synchronization problems
- Consider P_1 and P_2 that require S_1 to happen before S_2
Create a semaphore “**synch**” initialized to 0
P1:
 S_1 ;
 signal(synch);
P2:
 wait(synch);
 S_2 ;
- Can implement a counting semaphore S as a binary semaphore

- Must guarantee that no two processes can execute the **wait()** and **signal()** on the same semaphore at the same time
- Thus, the implementation becomes the critical section problem where the **wait** and **signal** code are placed in the critical section
 - Could now have **busy waiting** in critical section implementation
 - 4 But implementation code is short
 - 4 Little busy waiting if critical section rarely occupied
- Note that applications may spend lots of time in critical sections and therefore this is not a good solution

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Semaphore Implementation with no Busy waiting



- With each semaphore there is an associated waiting queue
- Each entry in a waiting queue has two data items:
 - value (of type integer)
 - pointer to next record in the list
- Two operations:
 - **block** – place the process invoking the operation on the appropriate waiting queue
 - **wakeup** – remove one of processes in the waiting queue and place it in the ready queue
- **typedef struct{**
int value;
struct process *list;
} semaphore;

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Semaphore Implementation with no Busy waiting (Cont.)

```
wait(semaphore *S) {  
    S->value--;  
    if (S->value < 0) {  
        add this process to S->list;  
        block();  
    }  
}  
  
signal(semaphore *S) {  
    S->value++;  
    if (S->value >= 0) {  
        remove a process P from S->list;  
        wakeup(P);  
    }  
}
```



- Incorrect use of semaphore operations:
 - signal (mutex) wait (mutex)
 - wait (mutex) ... wait (mutex)
 - Omitting of wait (mutex) or signal (mutex) (or both)
- Deadlock and starvation are possible.
- **Deadlock** – two or more processes are waiting indefinitely for an event that can be caused by only one of the waiting processes
- Let **S** and **Q** be two semaphores initialized to 1

P_0		P_1
	wait(S) ;	wait(Q) ;
	wait(Q) ;	wait(S) ;
...	...	
	signal(S) ;	signal(Q) ;
	signal(Q) ;	signal(S) ;

- **Starvation – indefinite blocking**
 - A process may never be removed from the semaphore queue in which it is suspended
- **Priority Inversion** – Scheduling problem when lower-priority process holds a lock needed by higher-priority process
 - Solved via **priority-inheritance protocol**

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Priority Inheritance Protocol

- When several tasks are waiting for the same critical resource, the task which is currently holding this critical resource is given the highest priority among all the tasks which are waiting for the same critical resource.
- Now after the lower priority task having the critical resource is given the highest priority then the intermediate priority tasks can not preempt this task. This helps in avoiding priority inversion.
- When the task which is given the highest priority among all tasks, finishes the job and releases the critical resource then it gets back to its original priority value (which may be less or equal).
- It allows the different priority tasks to share the critical resources.



- Consider the scenario with three processes **P1**, **P2**, and **P3**.
 - **P1** has the highest priority, **P2** the next highest, and **P3** the lowest.
- Assume that **P3** is holding semaphore **S** and that **P1** is waiting for **S** to be released
- Assume that **P2** is assigned the CPU and preempts **P3**
 - **P3** is still holding semaphore **S**
 - **P1** is waiting for **S** to be released
- What has happened is that **P2** - a process with a lower priority than **P1** - has indirectly prevented **P3** from gaining access to the resource.
- To prevent this from occurring, a **priority inheritance protocol** is used.



THANK YOU

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UE24CS242B

Classic problems of Synchronization

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Slides Credits for all the PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:
 1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
 2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
 3. Some presentation transcripts from A. Frank – P. Weisberg
 4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

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Classical problems of Synchronization

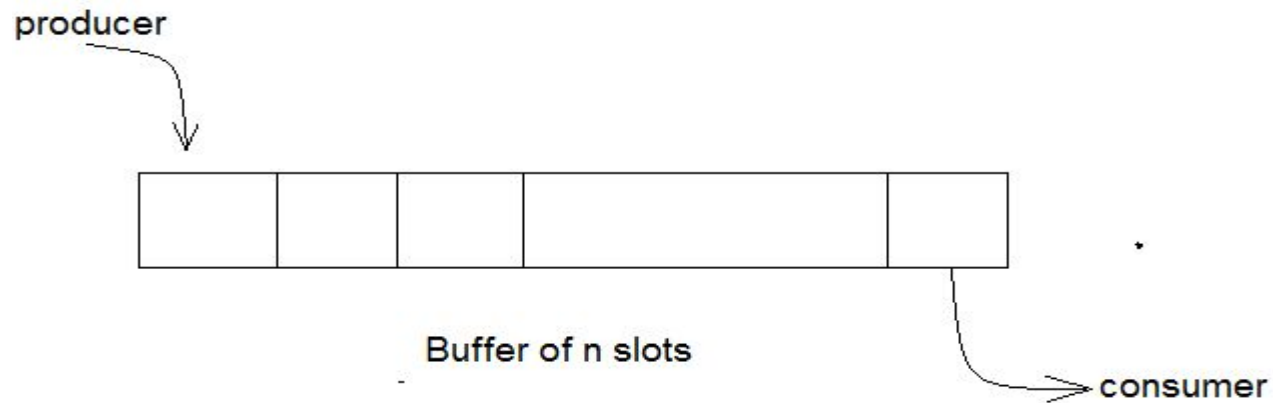
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Bounded-Buffer Problem

- n buffers, each can hold one item
- Semaphore **mutex** initialized to the value 1
- Semaphore **full** initialized to the value 0
- Semaphore **empty** initialized to the value n



- The structure of the producer process

```
do {  
    ...  
    /* produce an item in next_produced */  
    ...  
    wait(empty); //wait until empty > 0 and then decrement 'empty'  
    wait(mutex); //acquire lock  
    ...  
    /* add next produced to the buffer */  
    ...  
    signal(mutex); //release a lock  
    signal(full); //increment full  
} while (true);
```

```
❑ The structure of the consumer process
do {
    wait(full); // wait until full > 0 and then decrement 'full'
    wait(mutex); // acquire the lock
    ...
    /* remove an item from buffer to next_consumed */
    ...
    signal(mutex); // release the lock
    signal(empty); // increment 'empty'
    ...
    /* consume the item in next consumed */
    ...
} while (true);
```


The Problem Statement

There is a shared resource which should be accessed by multiple processes. There are two types of processes in this context. They are **reader** and **writer**. Any number of **readers** can read from the shared resource simultaneously, but only one **writer** can write to the shared resource. When a **writer** is writing data to the resource, no other process can access the resource. A **writer** cannot write to the resource if there are non zero number of readers accessing the resource at that time.

- **solution**
- Several variations of how readers and writers are considered – all involve some form of priorities
- Shared Data
 - Data set
 - Semaphore **rw_mutex** initialized to 1(semaphore)
 - Semaphore **mutex** initialized to 1 (mutex)
 - Integer **read_count** initialized to 0

- The structure of a writer process

```
do {  
    wait(rw_mutex);  
  
    ...  
    /* writing is performed */  
  
    ...  
    signal(rw_mutex);  
} while (true);
```

- The structure of a reader process

```
do {  
    wait(mutex);  
    read_count++;  
    if (read_count == 1)  
        wait(rw_mutex);  
    signal(mutex);  
  
    ...  
    /* reading is performed */  
    ...  
    wait(mutex);  
    read_count--;  
    if (read_count == 0)  
        signal(rw_mutex);  
    signal(mutex);  
} while (true);
```

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Readers-Writers Problem Variations

- **First** variation – no reader kept waiting unless writer has permission to use shared object
- **Second** variation – once writer is ready, it performs the write ASAP
- Both may have starvation leading to even more variations
- Problem is solved on some systems by kernel providing reader-writer locks



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Dining-Philosophers Problem

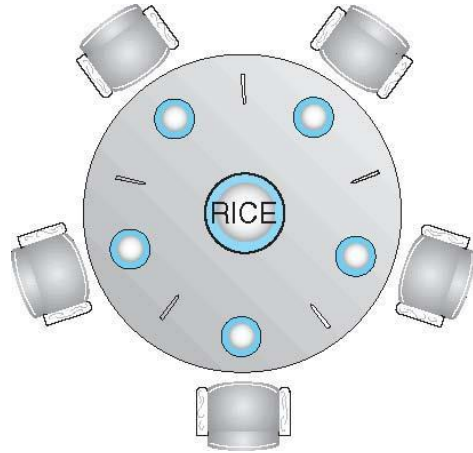


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Dining-Philosophers Problem



- Philosophers spend their lives alternating thinking and eating
- Don't interact with their neighbors, occasionally try to pick up 2 chopsticks (one at a time) to eat from bowl
 - Need both to eat, then release both when done
- In the case of 5 philosophers
 - Shared data
 - Bowl of rice (data set)
 - Semaphore **chopstick** [5] initialized to 1

- The structure of Philosopher i :
 do {
 wait (chopstick[i]);
 wait (chopstick[(i + 1) % 5]);
 // eat
 signal (chopstick[i]);
 signal (chopstick[(i + 1) % 5]);
 // think
 } while (TRUE);
- What is the problem with this algorithm?

- Deadlock handling
 - Allow at most 4 philosophers to be sitting simultaneously at the table.
 - Allow a philosopher to pick up the chopsticks only if both are available (picking must be done in a critical section).
 - Use an asymmetric solution -- an odd-numbered philosopher picks up first the left chopstick and then the right chopstick. Even-numbered philosopher picks up first the right chopstick and then the left chopstick.



THANK YOU

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